

NADOR AL AROUI, Morocco

WMO: 603400

Lat: 34.983N

Lon: 3.017W

Elev: 177

StdP: 99.22

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.0	3.5	-2.2	3.2	11.1	-0.3	3.8	9.6	12.4	14.7	10.7	14.8	1.7	260	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.2	34.7	20.4	32.9	20.4	31.2	20.8	25.5	28.6	24.6	28.0	23.7	27.5	4.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	20.2	27.3	23.6	18.8	26.6	22.6	17.7	26.0	79.3	28.1	75.2	28.3	71.9	27.6	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	8.9	7.8	0.2	38.1	2.3	2.4	-1.4	39.8	-2.7	41.2	-4.0	42.6	-5.7	44.3		
			-0.6	26.1	2.4	2.3	-2.3	27.8	-3.7	29.2	-5.0	30.5	-6.7	32.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	11.7	12.5	14.1	15.9	18.7	22.3	25.1	25.8	22.9	19.8	15.5	12.6
	DBStd	5.39	2.63	2.63	2.30	2.01	2.14	2.16	2.06	1.99	2.14	2.40	2.92	2.51
	HDD10.0	26	12	7	1	0	0	0	0	0	0	1	6	
	HDD18.3	881	206	162	133	76	20	1	0	1	13	94	177	
	CDD10.0	2992	65	78	128	178	270	368	469	491	388	305	164	88
	CDD18.3	806	0	1	1	4	32	119	211	232	138	59	8	1
	CDH23.3	6815	1	4	19	53	249	914	1994	2270	959	302	47	3
CDH26.7	2388	0	1	2	8	53	277	809	909	259	62	7	0	
Wind	WSAvg	3.1	2.8	3.3	3.4	3.4	3.3	3.4	3.3	3.3	3.2	2.7	3.0	2.5
Precipitation	PrecAvg	343	39	36	38	34	20	6	2	6	21	39	51	50
	PrecMax	743	157	117	124	123	48	53	6	37	78	216	156	156
	PrecMin	187	0	0	0	1	1	0	1	2	2	1	0	1
	PrecStd	122	37	27	33	28	15	12	2	7	19	43	46	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	23.0	25.9	27.8	30.9	33.9	37.2	37.1	33.1	30.9	27.2	22.8
		MCWB	13.7	14.5	16.9	16.7	18.6	20.9	21.5	21.2	20.2	19.4	16.3	16.0
	2%	DB	20.1	20.7	23.0	24.7	28.0	31.5	34.9	34.8	31.0	28.2	24.2	20.8
		MCWB	13.6	14.2	15.1	16.1	17.6	20.0	20.5	20.3	20.7	19.5	16.7	14.6
	5%	DB	18.5	19.0	21.0	22.9	26.0	29.9	32.9	32.9	29.3	26.2	22.4	19.3
		MCWB	13.0	13.5	14.6	15.7	17.3	20.0	20.2	20.8	20.9	19.0	16.3	14.1
	10%	DB	17.0	17.6	19.2	21.2	24.2	28.1	31.0	31.1	28.0	24.8	20.9	18.0
		MCWB	12.5	13.2	14.1	15.1	17.0	19.5	20.5	21.6	20.7	18.9	15.9	13.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.1	16.6	18.3	18.7	21.5	24.9	26.5	27.1	26.0	23.2	20.0	18.0
		MCDB	19.8	20.2	22.7	23.6	26.1	30.2	31.2	30.0	28.8	27.4	23.6	21.0
	2%	WB	14.9	15.4	16.6	17.5	20.1	23.0	25.0	25.9	24.2	21.6	18.6	16.2
		MCDB	17.9	18.4	20.5	22.2	24.2	26.9	28.2	28.3	27.4	24.5	22.1	18.7
	5%	WB	14.0	14.6	15.6	16.7	19.2	22.0	24.1	24.9	22.9	20.7	17.6	15.1
		MCDB	16.9	17.5	19.2	20.8	23.0	26.0	27.8	28.2	26.5	23.6	20.7	18.2
	10%	WB	13.1	13.9	14.9	16.0	18.3	21.2	23.3	24.1	22.1	20.0	16.7	14.1
		MCDB	16.2	16.8	18.0	19.7	22.3	25.4	27.1	27.7	25.5	22.9	19.8	17.3
Mean Daily Temperature Range	5% DB	MDBR	10.7	10.0	10.0	10.5	10.4	10.2	10.2	10.2	9.9	10.2	10.1	10.6
		MCDBR	12.9	12.5	14.4	14.1	14.5	13.8	14.6	14.3	13.0	13.1	12.4	12.1
		MCWBR	7.5	7.4	8.2	7.2	6.8	5.8	5.8	6.3	6.1	6.3	6.6	7.3
	5% WB	MCDBR	10.6	9.8	11.2	11.0	10.2	9.8	8.8	8.6	9.0	9.2	9.5	10.1
		MCWBR	6.9	6.5	7.3	6.7	6.0	5.5	4.9	4.9	5.4	5.3	5.9	7.0
Clear-Sky Solar Irradiance	taub	0.316	0.346	0.403	0.414	0.447	0.481	0.521	0.505	0.460	0.410	0.356	0.324	
	taud	2.493	2.406	2.225	2.215	2.149	2.053	1.943	2.001	2.137	2.271	2.417	2.493	
	Ebn at Noon	891	896	864	872	844	812	776	781	798	810	830	856	
	Edh at Noon	84	102	133	141	153	168	186	173	144	116	90	80	
All-Sky Solar Radiation	RadAvg	2.86	3.74	4.90	5.95	6.83	7.58	7.43	6.64	5.39	4.08	3.01	2.60	
	RadStd	0.23	0.31	0.39	0.42	0.50	0.23	0.18	0.23	0.23	0.26	0.23	0.19	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page